

Extrinsic Calibration of 2-D Lidars Using Two Orthogonal Planes

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Abstract—This paper describes a new methodology for estimating the relative pose between two 2-D lidars. Scanned points of 2-D lidars do not have enough feature information for correspondence matching. For this reason, additional image sensors or artificial landmarks at known locations have been used to find the relative pose. We propose a novel method of estimating the relative pose between 2-D lidars without any additional sensors or artificial landmarks. By scanning two orthogonal planes, we utilize the coplanarity of the scan points on each plane and the orthogonality of the plane normals. Even if we capture planes which are not exactly orthogonal, the method provides good results using nonlinear optimization. Experiments with both synthetic and real data show the validity of the proposed method. We also derive two degenerate cases: one related to plane poses, and the other caused by the relative pose. To the best of our knowledge, this study provides the first solution for the problem.

Index Terms—Calibration, orthogonal planes, 2-D lidar.

I. INTRODUCTION

LIDARS are one of the most important sensors in many robotic applications, mainly because of their accuracy and robustness of their measured distances. Wolf and Sukhatme [1] showed the usefulness of lidars for the successful simultaneous localization and mapping (SLAM) of a 2-D environment. With the evolution of robotic applications from 2-D to 3-D, the demand for 3-D applications that use range information has been growing. Many works [2]–[4] have presented 6-D SLAM using a camera and a lidar in outdoor environments. An unmanned autonomous vehicle is usually equipped with many sensors for such tasks. In the 2005 DARPA Grand Challenge and in the 2007 Urban Challenge, Thrun *et al.* [5], [6] successfully achieved important tasks such as vehicle motion estimation and target

detection using several lidars, cameras, and a 3-D lidar. Some researchers have proposed methods of reconstructing 3-D structures using lidars. Fentanes *et al.* [7] proposed an algorithm of navigation and 3-D reconstruction using cameras, GPS, IMU, and two 2-D lidars. Bok *et al.* [8] showed large-scale 3-D reconstruction in an outdoor environment using two 2-D lidars and two cameras.

In these systems using multiple sensors, registration or extrinsic calibration of the sensors is critical; thus, many efforts have been made to ensure accurate registration between various sensors. However, there have been surprisingly few works on calibrating multiple 2-D lidars due to the difficulties in searching for correspondences among scan data. One of the most popular methods developed so far to calibrate multiple 2-D lidars is to use an additional camera as a bridge. The extrinsic parameter between a camera and a 2-D lidar can be computed using geometric constraints. Zhang and Pless [9] proposed a calibration method when a planar pattern is posed in the both views of the sensors. They used a constraint in which the camera observations and the lidar scan data are placed on the same plane. Bok *et al.* [10] calculated the extrinsic parameters of the sensors without overlap. They utilized a plane or a line that is parallel to one of the three axes of the pattern coordinate system. Ha [11] calibrated the sensors using plane structures with a triangular hole, not geometric constraints. He computed 3-D–3-D correspondences between the camera and the lidar using range discontinuities on the proposed calibration structure.

Artificial landmarks that are visible to lidars can also be used to calculate correspondences among scan data. Gao and Spletzer [12] and Underwood *et al.* [13] proposed extrinsic calibration methods using a vertical pole with reflective targets. The targets are attached at known positions so that a user can determine the exact pose between the lidar and a mobile platform.

Different from 2-D lidar data, point clouds scanned using 3-D lidars can be aligned using geometric features. Sheehan *et al.* [14] proposed a 3-D scanner that consists of three 2-D lidars mounted on a spinning plate. They adopted Renyi quadratic entropy to measure the registration error between the 3-D point clouds. Using Renyi entropy and additional information from an inertial navigation system, Maddern *et al.* [15] calculated the poses of two 2-D lidars and a 3-D scanner proposed in [14] with respect to a vehicle coordinate system. Levinson and Thrun [16] proposed a calibration method that calculates both the intrinsic and extrinsic parameters of a multibeam lidar (Velodyne HDL-64E). They iteratively minimized errors among surface normals obtained from local neighborhoods. Recently, He *et al.* [17] demonstrated a calibration method for a multiple 2-D

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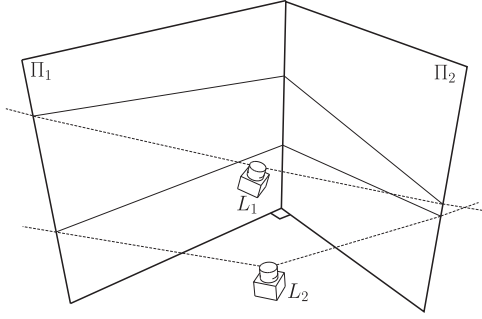


Fig. 1. Setup of the proposed extrinsic calibration method. We calculate a relative pose from L_2 to L_1 using two orthogonal planes.

lidar system. They accumulated scan data using the motion of a platform in an underground parking lot. They registered plane features extracted manually from point clouds.

We propose a new method for registering two 2-D lidars directly without any extra sensors or known motion information. Direct calibration between 2-D lidars is useful for various systems that utilize two or more 2-D lidars. Tiny systems such as UAVs and portable devices are good examples because it is hard to mount additional sensors on such systems. The proposed method is also useful for designing 3-D sensors that consist of rotating modules and 2-D lidars. Even if a system can be calibrated using conventional methods, the proposed method provides a good constraint for system calibration. The proposed method uses only scan data on two orthogonal planes, as shown in Fig. 1. At each scan, each plane has two scan lines from the two lidars; the coplanarity of the scan points provides a linear constraint to an unknown vector composed by the extrinsic parameters. The extrinsic parameters are extracted from the unknown vector using the orthogonality between the two planes. To refine the solution, we apply a nonlinear optimization that minimizes the squared sum of the distances between the scanned data and the planes. Even if the scanned planes are not exactly orthogonal, the proposed method gives good results when using the optimization process. Two degenerate cases (critical capture motion and relative pose) are found in which the calibration solution fails.

This paper extends our previous work [18] by improving algorithms to derive a linear constraint equation and determine a scale value. We also analyze degenerate cases of the proposed method. Additional synthetic and real experiments are given to verify the accuracy and robustness of the proposed method.

II. EXTRINSIC CALIBRATION OF TWO SINGLE-LINE LASER SENSORS

This section provides how to calibrate the extrinsic parameters between two 2-D lidar sensors. We will derive a linear constraint on the relative pose between two lidars and explain how to estimate relative pose from the linearly constrained solution using another constraint: orthogonality. Refinement of the solution with nonlinear optimization will also be discussed in this section.

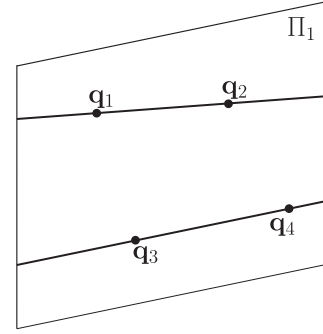


Fig. 2. Coplanarity of the scan points provides a linear constraint to the unknown vector composed by the extrinsic parameters.

Let us assume that we have two lidars L_1 and L_2 , and a plane Π_1 , as shown in Fig. 2. Each lidar scan plane intersects Π_1 and for L_1 ; there are two points $\mathbf{q}_1$ and $\mathbf{q}_2$ defining the intersecting line in 3-D. Similarly, for L_2 , we can define two points $\mathbf{q}_3$ and $\mathbf{q}_4$. It should be noted that all the points are represented in a common coordinate system, which is the coordinate system of L_1 . The points represented in the coordinate system of L_1 and those in the L_2 coordinate system will be denoted by $\mathbf{q}_n = [x_n \ y_n \ z_n]^\top$ and $\mathbf{q}'_n = [x'_n \ y'_n \ z'_n]^\top$, respectively. These points can be transformed using

$$\mathbf{q}_n = \mathbf{R}\mathbf{q}'_n + \mathbf{t} \quad (1)$$

where $\mathbf{R}$ and $\mathbf{t}$ are a rotation matrix and a translation vector that transforms points from L_2 coordinate to L_1 coordinate. The element of $\mathbf{R}$ at the i th row and the j th column is denoted by r_{ij} ; we also let $\mathbf{t} = [t_x \ t_y \ t_z]^\top$. The difference between the two points will also be denoted by $\Delta\mathbf{q}_{mn} = \mathbf{q}_n - \mathbf{q}_m = [\Delta x_{mn} \ \Delta y_{mn} \ \Delta z_{mn}]^\top$ and $\Delta\mathbf{q}'_{mn} = \mathbf{q}'_n - \mathbf{q}'_m = [\Delta x'_{mn} \ \Delta y'_{mn} \ \Delta z'_{mn}]^\top$.

A 3-D line $\mathbf{l}_{mn}$ through $\mathbf{q}_n$ and $\mathbf{q}_m$ is represented using a Plücker coordinate as

$$\mathbf{l}_{mn} = \langle \mathbf{d}_{mn}, \mathbf{m}_{mn} \rangle = \langle \Delta\mathbf{q}_{mn}, [\mathbf{q}_n]_\times \mathbf{q}_m \rangle \quad (2)$$

which is a six-vector. The first three-vector $\Delta\mathbf{q}_{mn}$ represents its *direction*, and the latter three-vector $\mathbf{q}_n \times \mathbf{q}_m$ is called its *moment*.

A line $\mathbf{l}'_{mn}$ is transformed into another coordinate system with a rotation $\mathbf{R}$ and a translation $\mathbf{t}$ using a *Line Motion Matrix* [19], as

$$\mathbf{l}_{mn} = \begin{bmatrix} \mathbf{R} & \mathbf{0}_{3 \times 3} \\ [\mathbf{t}]_\times \mathbf{R} & \mathbf{R} \end{bmatrix} \mathbf{l}'_{mn} \quad (3)$$

where $[\cdot]_\times$ denotes the skew symmetric matrix.

A. Linear Constraint of Coplanar Lidar Scan Points

When two 2-D scan lidars L_1 and L_2 scan a single plane Π_1 , as shown in Fig. 2, the lines $\mathbf{l}_{21}$ and $\mathbf{l}_{43}$ should be coplanar. It derives

$$\begin{bmatrix} \Delta\mathbf{q}_{21} \\ [\mathbf{q}_1]_\times \mathbf{q}_2 \end{bmatrix}^\top \begin{bmatrix} \mathbf{0} & \mathbf{I}_{3 \times 3} \\ \mathbf{I}_{3 \times 3} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \Delta\mathbf{q}_{43} \\ [\mathbf{q}_3]_\times \mathbf{q}_4 \end{bmatrix} = 0 \quad (4)$$

and by applying the line motion matrix to the lidar L_2 , we have

$$\begin{bmatrix} \Delta \mathbf{q}_{21} \\ [\mathbf{q}_1]_{\times} \mathbf{q}_2 \end{bmatrix}^{\top} \begin{bmatrix} [\mathbf{t}]_{\times} \mathbf{R} & \mathbf{R} \\ \mathbf{R} & \mathbf{0}_{3 \times 3} \end{bmatrix} \begin{bmatrix} \Delta \mathbf{q}'_{43} \\ [\mathbf{q}'_3]_{\times} \mathbf{q}'_4 \end{bmatrix} = 0 \quad (5)$$

which is the same as the generalized epipolar constraint of multiple cameras [20]. We refer to the 6×6 matrix as a *Lidar Essential Matrix*.

One conventional method to estimate the relative motion of a set of multiple cameras is to estimate the generalized essential matrix by providing a sufficient number of correspondences, and then to expose the rotation constraints on the rotation part. However, in the single-line lidar case, this cannot be directly achieved, because each lidar scans points only on a single plane, i.e., $y_n = y'_n = 0$ for all n .

The single-line scanning condition $y_n = 0$ makes meaningful only two components x and z for a point $\mathbf{q}_n$ and one component y for a cross product $[\mathbf{q}_n]_{\times} \mathbf{q}_m$. To reflect this condition in (5), we adopt two row sampling matrices $\mathbf{M}_1$ and $\mathbf{M}_2$, defined as

$$\mathbf{M}_1 \equiv \begin{bmatrix} \mathbf{e}_1^{\top} \\ \mathbf{e}_3^{\top} \end{bmatrix}, \quad \mathbf{M}_2 \equiv \mathbf{e}_2^{\top} \quad (6)$$

where $\mathbf{e}_1 = [1 \ 0 \ 0]^{\top}$, $\mathbf{e}_2 = [0 \ 1 \ 0]^{\top}$, and $\mathbf{e}_3 = [0 \ 0 \ 1]^{\top}$. Obviously, $\mathbf{M}_1$ is for directions, and $\mathbf{M}_2$ is for moments. A row sampled 3-vector $\mathbf{L}_{nm}$ of the stacked vector is made by stacking $\mathbf{M}_1$ and $\mathbf{M}_2$ as

$$\mathbf{L}_{nm} = \mathbf{M}_{3 \times 6} \begin{bmatrix} \Delta \mathbf{q}_{mn} \\ [\mathbf{q}_n]_{\times} \mathbf{q}_m \end{bmatrix}, \quad \text{where } \mathbf{M}_{3 \times 6} \equiv \begin{bmatrix} \mathbf{M}_1 & \mathbf{0}_{2 \times 3} \\ \mathbf{0}_{1 \times 3} & \mathbf{M}_2 \end{bmatrix} \quad (7)$$

The lidar essential matrix is also row and column sampled, and its reduced form is given as

$$\mathbf{E}_L = \mathbf{M}_{3 \times 6} \begin{bmatrix} [\mathbf{t}]_{\times} \mathbf{R} & \mathbf{R} \\ \mathbf{R} & \mathbf{0}_{3 \times 3} \end{bmatrix} \mathbf{M}_{3 \times 6}^{\top}. \quad (8)$$

We call $\mathbf{E}_L$ a *reduced lidar essential matrix*, which is a 3×3 matrix.

In summary, if two 2-D lidars scan a single plane, we have a single bilinear constraint as

$$\mathbf{L}_{12}^{\top} \mathbf{E}_L \mathbf{L}'_{34} = 0. \quad (9)$$

The reduced lidar essential matrix $\mathbf{E}_L$ is defined with parameters of $\mathbf{R}$ and $\mathbf{t}$ as

$$\mathbf{E}_L = \begin{bmatrix} r_{31}t_y - r_{21}t_z & r_{33}t_y - r_{23}t_z & r_{12} \\ r_{21}t_x - r_{11}t_y & r_{23}t_x - r_{13}t_y & r_{32} \\ r_{21} & r_{23} & 0 \end{bmatrix} \quad (10)$$

and the bidirectional form in (9) is expanded as

$$\begin{bmatrix} \Delta x_{21}(z'_3x'_4 - x'_3z'_4) \\ \Delta z_{21}(z'_3x'_4 - x'_3z'_4) \\ \Delta x'_{43}(z_1x_2 - x_1z_2) \\ \Delta z'_{43}(z_1x_2 - x_1z_2) \\ \Delta x_{21}\Delta x'_{43} \\ \Delta x_{21}\Delta z'_{43} \\ \Delta z_{21}\Delta x'_{43} \\ \Delta z_{21}\Delta z'_{43} \end{bmatrix}^{\top} \begin{bmatrix} r_{12} \\ r_{32} \\ r_{21} \\ r_{23} \\ r_{31}t_y - r_{21}t_z \\ r_{33}t_y - r_{23}t_z \\ r_{21}t_x - r_{11}t_y \\ r_{23}t_x - r_{13}t_y \end{bmatrix} = 0. \quad (11)$$

By stacking the measurement vectors, a linear system $\mathbf{A}\mathbf{x} = 0$ is built. At least seven different poses of planes are required to solve the equation via singular value decomposition (SVD).

Once the reduced lidar essential matrix is estimated, the pose parameters should be extracted from the matrix. For a conventional generalized essential matrix, the orthonormal constraint of the rotation matrix is just enough to uniquely determine the pose parameters from the estimated matrix. However, the single line scanning condition degenerates the process. In Appendix A, we show that it is not possible to determine the pose parameters uniquely from a reduced lidar essential matrix. In the following, we develop a method to determine the pose parameters using another independent constraint.

B. Pose Parameters From Least-Squares Solution

We formulate pose parameters from the least-squares solution. Let $\mathbf{v}$ be the right singular vector of $\mathbf{A}$ corresponding to its smallest singular value, and v_i be the i th element of $\mathbf{v}$. The solution $\mathbf{v}$ is obtained up to scale: the final solution is $\lambda\mathbf{v}$ with an unknown scale factor λ . The solution vector $\lambda\mathbf{v}$ is composed of the pose parameters $\mathbf{R}$ and $\mathbf{t}$, as shown in (11). We extract $\mathbf{R}$ from $\mathbf{v}$ using its orthonormality

$$\mathbf{R} = \begin{bmatrix} -k_1r_{22} \pm k_2 & \lambda v_1 & -k_4r_{22} \mp k_3 \\ \lambda v_3 & r_{22} & \lambda v_4 \\ -k_3r_{22} \mp k_4 & \lambda v_2 & -k_2r_{22} \pm k_1 \end{bmatrix} \quad (12)$$

where we defined that as

$$\begin{aligned} r_{22} &\equiv \pm \sqrt{1 - \lambda^2 k_0} \\ k_0 &\equiv v_1^2 + v_2^2 \\ k_1 &\equiv \frac{v_1 v_3}{v_1^2 + v_2^2} \\ k_2 &\equiv \frac{v_2 v_4}{v_1^2 + v_2^2} \\ k_3 &\equiv \frac{v_2 v_3}{v_1^2 + v_2^2} \\ k_4 &\equiv \frac{v_1 v_4}{v_1^2 + v_2^2}. \end{aligned} \quad (13)$$

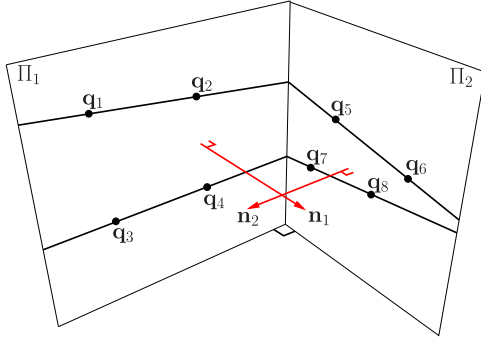


Fig. 3. Unknown scale factors in (12) and (14) are determined by another constraint: orthogonality between two scene planes, which completes the relative pose estimation.

Note that the constants k_n are determined only with the known vector $\mathbf{v}$. A detailed derivation of the rotation parameters is given in Appendix B.

The translation vector $\mathbf{t}$ is obtained from the bottom half of (11) as

$$\begin{bmatrix} 0 & r_{31} & -r_{21} & -\lambda v_5 \\ 0 & r_{33} & -r_{23} & -\lambda v_6 \\ r_{21} & -r_{11} & 0 & -\lambda v_7 \\ r_{23} & -r_{13} & 0 & -\lambda v_8 \end{bmatrix} \begin{bmatrix} t_x \\ t_y \\ t_z \\ 1 \end{bmatrix} = 0. \quad (14)$$

Because we have four equations with three unknowns, the translation vector is obtained using a least-squares method with SVD.

C. Scale Determination Using Orthogonality

An additional equation is required to determine the scale λ . In this paper, we utilize a pair of perpendicular planes. Scan data from both planes can be used to compute $\mathbf{v}$ because the planes share common values of $\mathbf{R}$ and $\mathbf{t}$. Normal vectors of the planes are functions of λ . The scale is determined using the orthogonal property of the vectors. Let $\mathbf{n}_1$ and $\mathbf{n}_2$ be the normal vectors of two orthogonal planes Π_1 and Π_2 , respectively, as shown in Fig. 3.

Scanning the planes, two lines can be extracted from the range data. Lines on planes Π_1 and Π_2 scanned by L_1 are defined by $(\mathbf{q}_1, \mathbf{q}_2)$ and $(\mathbf{q}_5, \mathbf{q}_6)$, respectively, and those scanned by L_2 are defined by $(\mathbf{q}_3, \mathbf{q}_4)$ and $(\mathbf{q}_7, \mathbf{q}_8)$, respectively, as shown in Fig. 3. Normal vectors $\mathbf{n}_1$ and $\mathbf{n}_2$ are computed using these points

$$\mathbf{n}_1 = \Delta \mathbf{q}_{21} \times \Delta \mathbf{q}_{43} = \begin{bmatrix} \Delta x_{21} \\ 0 \\ \Delta z_{21} \end{bmatrix} \times \begin{bmatrix} r_{11} \Delta x'_{43} + r_{13} \Delta z'_{43} \\ r_{21} \Delta x'_{43} + r_{23} \Delta z'_{43} \\ r_{31} \Delta x'_{43} + r_{33} \Delta z'_{43} \end{bmatrix} \quad (15)$$

$$\mathbf{n}_2 = \Delta \mathbf{q}_{56} \times \Delta \mathbf{q}_{78} = \begin{bmatrix} \Delta x_{56} \\ 0 \\ \Delta z_{56} \end{bmatrix} \times \begin{bmatrix} r_{11} \Delta x'_{78} + r_{13} \Delta z'_{78} \\ r_{21} \Delta x'_{78} + r_{23} \Delta z'_{78} \\ r_{31} \Delta x'_{78} + r_{33} \Delta z'_{78} \end{bmatrix}. \quad (16)$$

The dot product of the normal vectors must be equal to zero

$$\mathbf{n}_1 \cdot \mathbf{n}_2 = \begin{bmatrix} \lambda k_5 \\ k_6 r_{22} + k_7 \\ \lambda k_8 \end{bmatrix} \cdot \begin{bmatrix} \lambda k_9 \\ k_{10} r_{22} + k_{11} \\ \lambda k_{12} \end{bmatrix} = 0 \quad (17)$$

where we defined that

$$\begin{aligned} k_5 &\equiv -v_3 \Delta z_{21} \Delta x'_{43} - v_4 \Delta z_{21} \Delta z'_{43} \\ k_6 &\equiv -k_1 \Delta z_{21} \Delta x'_{43} - k_4 \Delta z_{21} \Delta z'_{43} \\ &\quad + k_3 \Delta x_{21} \Delta x'_{43} + k_2 \Delta x_{21} \Delta z'_{43} \\ k_7 &\equiv \pm k_2 \Delta z_{21} \Delta x'_{43} \mp k_3 \Delta z_{21} \Delta z'_{43} \\ &\quad \pm k_4 \Delta x_{21} \Delta x'_{43} \mp k_1 \Delta x_{21} \Delta z'_{43} \\ k_8 &\equiv v_3 \Delta x_{21} \Delta x'_{43} + v_4 \Delta x_{21} \Delta z'_{43} \\ k_9 &\equiv -v_3 \Delta z_{56} \Delta x'_{78} - v_4 \Delta z_{56} \Delta z'_{78} \\ k_{10} &\equiv -k_1 \Delta z_{56} \Delta x'_{78} - k_4 \Delta z_{56} \Delta z'_{78} \\ &\quad + k_3 \Delta x_{56} \Delta x'_{78} + k_2 \Delta x_{56} \Delta z'_{78} \\ k_{11} &\equiv \pm k_2 \Delta z_{56} \Delta x'_{78} \mp k_3 \Delta z_{56} \Delta z'_{78} \\ &\quad \pm k_4 \Delta x_{56} \Delta x'_{78} \mp k_1 \Delta x_{56} \Delta z'_{78} \\ k_{12} &\equiv v_3 \Delta x_{56} \Delta x'_{78} + v_4 \Delta x_{56} \Delta z'_{78}. \end{aligned} \quad (18)$$

Substituting $r_{22} = \pm \sqrt{1 - \lambda^2 k_0}$, (17) is expressed as a fourth-order polynomial equation of λ :

$$k_{13}^2 \lambda^4 + (2k_{13}k_{15} + k_0 k_{14}^2) \lambda^2 - k_{14}^2 + k_{15}^2 = 0 \quad (19)$$

where we defined that

$$\begin{aligned} k_{13} &\equiv k_5 k_9 + k_8 k_{12} - k_0 k_6 k_{10} \\ k_{14} &\equiv k_6 k_{11} + k_7 k_{10} \\ k_{15} &\equiv k_6 k_{10} + k_7 k_{11}. \end{aligned} \quad (20)$$

Because the constant values with double signs in the same order, k_7 , k_{11} , and r_{22} , are squared or multiplied by each other in (19), only one quartic equation is derived from each scan.

The scale λ is computed by substituting $\mu = \lambda^2$, solving a quadratic equation and selecting positive solutions for μ :

$$f(\mu) = k_{13}^2 \mu^2 + (2k_{13}k_{15} + k_0 k_{14}^2) \mu - k_{14}^2 + k_{15}^2 = 0. \quad (21)$$

Next, we approximate the squared scale value μ' as a minimization of the squared summation of the $f(\mu)$ as

$$\mu' = \arg \min_{\mu} \sum_{i=1}^n f_i^2(\mu) \quad (22)$$

where n is the number of lidar scans that were used in the calibration process. The value of μ' is computed by solving a third-order polynomial equation derived by differentiating (22) in terms of μ :

$$\frac{\delta}{\delta \mu} \sum_{i=1}^n f_i^2(\mu) \big|_{\mu=\mu'} = 0. \quad (23)$$

Because the scale λ is replaced by $\mu = \lambda^2$, both signs of the final solution satisfy (19). The other solution with the opposite

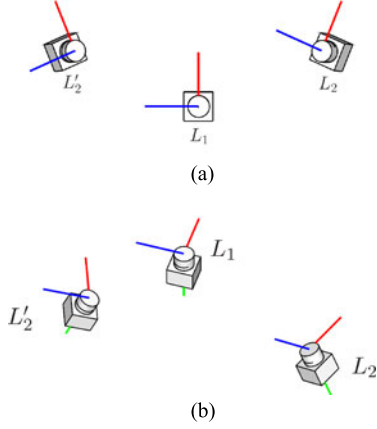


Fig. 4. Sensor configurations using the true scale value L_2 and the mirrored scale value L'_2 . Red, green, and blue lines indicate the x -, y -, and z -axes of each sensor coordinate system, respectively. It is easy to choose the true pose by comparing them with the actual sensor configuration. (a) Top view. (b) Side view.

sign is referred to as a “mirrored scale value” in the rest of this paper. Fig. 4 provides examples of two different sensor configurations derived from the solution pair. The solutions using the true and the mirrored scale value are denoted by L_2 and L'_2 , respectively. Both of the configurations satisfy all of the constraints proposed, and thus, it is not possible to choose the right solution between them. However, it is easy to choose the right one by comparing solutions to the actual sensor configuration, because the two possible poses have different signs in translation with respect to L_1 .

D. Nonlinear Optimization

The initial solution computed in Section II-C is refined via nonlinear optimization. We assume that the two planes are perpendicular. However, planes in real environments usually do not satisfy this assumption perfectly. We include the angle between the planes as an unknown parameter with an initial value of 90° . Moreover, we estimate the pose of the sensor rig in each scan with respect to the planes in the optimization. The initial values of the pose for each scan must be computed first.

The pose of the sensor rig with respect to the planes has five degrees of freedom because the translation along an intersection line of two planes cannot be estimated (i.e., a translation vector has two degrees of freedom). Without loss of generality, we set the intersection line of the two planes as the y -axis (i.e., both planes contain the y -axis), and the normal vector $\mathbf{n}_1$ of Π_1 as the x -axis of the world coordinate system. For each scan, data of L_2 are transformed into L_1 coordinate system and the normal vectors $\mathbf{n}_1$ and $\mathbf{n}_2$ of the two planes are computed by plane fitting. Since the normal vectors are computed up to scale, we fix their directions so that the signs of $\mathbf{n}_1 \cdot (\mathbf{q}_6 - \mathbf{q}_5)$ and $\mathbf{n}_2 \cdot (\mathbf{q}_1 - \mathbf{q}_2)$ must be positive, when the scanned surfaces are concave (see Fig. 3). After computing the normal vectors and their directions, we set three orthonormal vectors $\mathbf{w}_x$, $\mathbf{w}_y$, and $\mathbf{w}_z$ as the x -, y -, and z -axes of the world coordinate system,

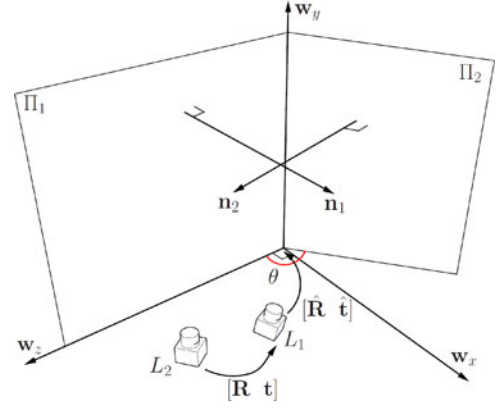


Fig. 5. Parameters for nonlinear optimization. The normal vector $\mathbf{n}_1$ of Π_1 , and the intersecting line of the two planes are selected to be the x - and y -axes, respectively, in the world coordinate system. The normal vector $\mathbf{n}_2$ of Π_2 is a function of θ , the angle between the two planes.

respectively

$$\mathbf{w}_x = \frac{\mathbf{n}_1}{\|\mathbf{n}_1\|} \quad (24)$$

$$\mathbf{w}_y = \frac{\mathbf{n}_2 \times \mathbf{n}_1}{\|\mathbf{n}_2 \times \mathbf{n}_1\|} \quad (25)$$

$$\mathbf{w}_z = \mathbf{w}_x \times \mathbf{w}_y. \quad (26)$$

Rotation part $\hat{\mathbf{R}}_i$ of the transformation from L_1 to the world coordinate system in the i th scan data is computed as

$$\hat{\mathbf{R}}_i = [\mathbf{w}_x \quad \mathbf{w}_y \quad \mathbf{w}_z]^\top. \quad (27)$$

We have mentioned that the translation vector $\hat{\mathbf{t}}_i$ has only two degrees of freedom. Since we set the intersection line as the y -axis, the y element of the translation can be any value. Without loss of generality, we set this value to zero. The other elements are computed by transforming the intersecting point $\mathbf{q}_{int}$ of the two lines $\overline{\mathbf{q}_1\mathbf{q}_2}$ and $\overline{\mathbf{q}_5\mathbf{q}_6}$ into the world coordinate system using the fact that $\mathbf{q}_{int}$ must lie on the y -axis:

$$\mathbf{q}_{int} \sim (\mathbf{q}_1 \times \mathbf{q}_2) \times (\mathbf{q}_5 \times \mathbf{q}_6) \quad (28)$$

$$\hat{\mathbf{t}}_i = [-\mathbf{w}_x^\top \mathbf{q}_{int} \quad 0 \quad -\mathbf{w}_z^\top \mathbf{q}_{int}]^\top. \quad (29)$$

In the optimization process, the angle θ between the two planes affects their normal vectors. In our implementation, we set $\hat{\mathbf{n}}_1$ equal to the x -axis and adjust $\hat{\mathbf{n}}_2$ using (31) (see Fig. 5)

$$\hat{\mathbf{n}}_1 = [1 \quad 0 \quad 0]^\top \quad (30)$$

$$\hat{\mathbf{n}}_2 = [\cos \theta \quad 0 \quad \sin \theta]^\top. \quad (31)$$

The cost function f for optimization is the squared sum of the Euclidean distances between the two planes and the scan data in the world coordinate system. We consider the unknown relative pose $[\mathbf{R} \quad \mathbf{t}]$, the lidar-to-world transformations $[\hat{\mathbf{R}} \quad \hat{\mathbf{t}}]$, and the angle θ between two planes as the variables of the optimization. The constant terms of the planes are equal to zero because the

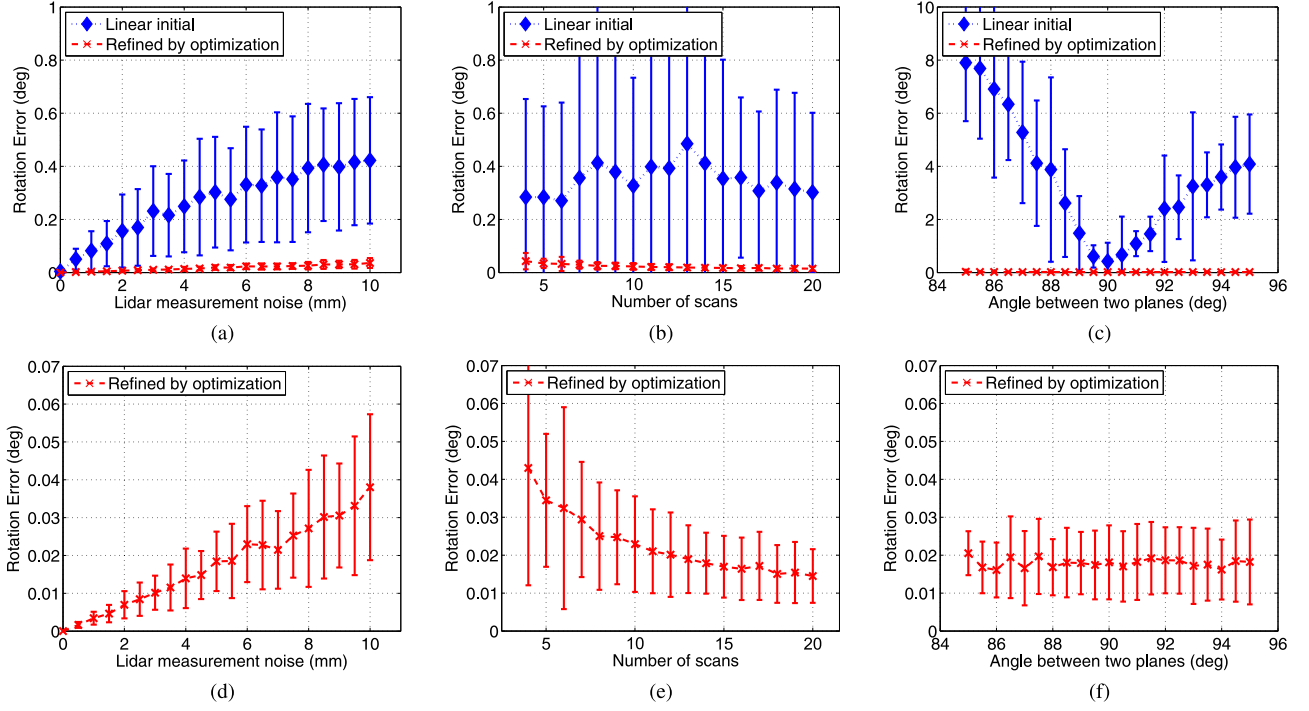


Fig. 6. Performance analysis using rotation errors. Initial (blue) and refined (red) results are analyzed in three aspects: 1. Gaussian noise in lidar measurement (a), (d); 2. The number of scans used in the estimation (b), (e); and 3. Errors in the orthogonality assumption (c), (f). (a)–(c) are comparison results between the initial and refined results. (d)–(f) show more details of the performance after the refinement.

origin of the world coordinate system is on both planes. Note that $\hat{\mathbf{n}}_2$ is a function of θ , as shown in (31). We adopt the Levenberg–Marquardt [21] algorithm for the nonlinear optimization

$$f(\mathbf{R}, \mathbf{t}, \hat{\mathbf{R}}_i, \hat{\mathbf{t}}_i, \theta) = \sum_{i=1}^n \left(\sum_{\mathbf{q} \in (L_1, \Pi_1)} (\hat{\mathbf{n}}_1^\top (\hat{\mathbf{R}}_i \mathbf{q} + \hat{\mathbf{t}}_i))^2 + \sum_{\mathbf{q} \in (L_1, \Pi_2)} (\hat{\mathbf{n}}_2^\top (\hat{\mathbf{R}}_i \mathbf{q} + \hat{\mathbf{t}}_i))^2 + \sum_{\mathbf{q}' \in (L_2, \Pi_1)} (\hat{\mathbf{n}}_1^\top (\hat{\mathbf{R}}_i (\mathbf{R} \mathbf{q}' + \mathbf{t}) + \hat{\mathbf{t}}_i))^2 + \sum_{\mathbf{q}' \in (L_2, \Pi_2)} (\hat{\mathbf{n}}_2^\top (\hat{\mathbf{R}}_i (\mathbf{R} \mathbf{q}' + \mathbf{t}) + \hat{\mathbf{t}}_i))^2 \right). \quad (32)$$

The whole process of the proposed method is described in Algorithm 1.

III. EXPERIMENTAL RESULTS

A. Simulation Using Synthetic Data

We generated a set of synthetic data to verify the performance of the proposed algorithm. Two planes $x = 0$ and $z = 0$ in the world coordinate system were used as orthogonal planes scanned by lidars. The rotation and translation of the L_1 -to-world transformation were generated randomly within $(-30, 30)$ (degrees) and $(0.1, 0.5)$ (m), respectively. Elements of rotation and translation of the relative poses of the L_2 -to- L_1 were

Algorithm 1 Extrinsic calibration of 2-D laser sensors

INPUT: Points $\mathbf{q}_1$ to $\mathbf{q}_8$ extracted from n captured scans

OUTPUT: Extrinsic calibration parameters, $\mathbf{R}$ and $\mathbf{t}$

for $i = 1 \sim n$ **do**

$\mathbf{A}(i, :) =$ The leftmost term of (11) using $\mathbf{q}_1$ to $\mathbf{q}_4$

$\mathbf{A}(i + n, :) =$ The leftmost term of (11) using $\mathbf{q}_5$ to $\mathbf{q}_8$

$i = i + 1$

end for

$[\mathbf{U}, \mathbf{D}, \mathbf{V}] = \text{SVD}(\mathbf{A})$

$[v_1 \sim v_8] = \mathbf{V}(:, 8)$

for $i = 1 \sim n$ **do**

 Compute coefficients k_{13} to k_{15} from (20)

$i = i + 1$

end for

Compute the scale from (23)

Compute initial values of $\mathbf{R}$ and $\mathbf{t}$ using (12) and (14)

Refine the initial pose using (32) as a cost function

randomly generated in intervals of $(-120, 120)$ (degrees) and $(-0.5, 0.5)$ (m), respectively.

We analyzed three aspects of the proposed algorithm: performance for measurement noise, errors in orthogonality assumption of the two planes, and the number of utilized scans. The experimental results are shown in Fig. 6. The Gaussian noise of the lidars from 0 to 10(mm) with a step of 0.5(mm) is added to the range data used in Fig. 6(a). The number of scans used for the calibration is changed from 4 to 20, as shown in Fig. 6(b),

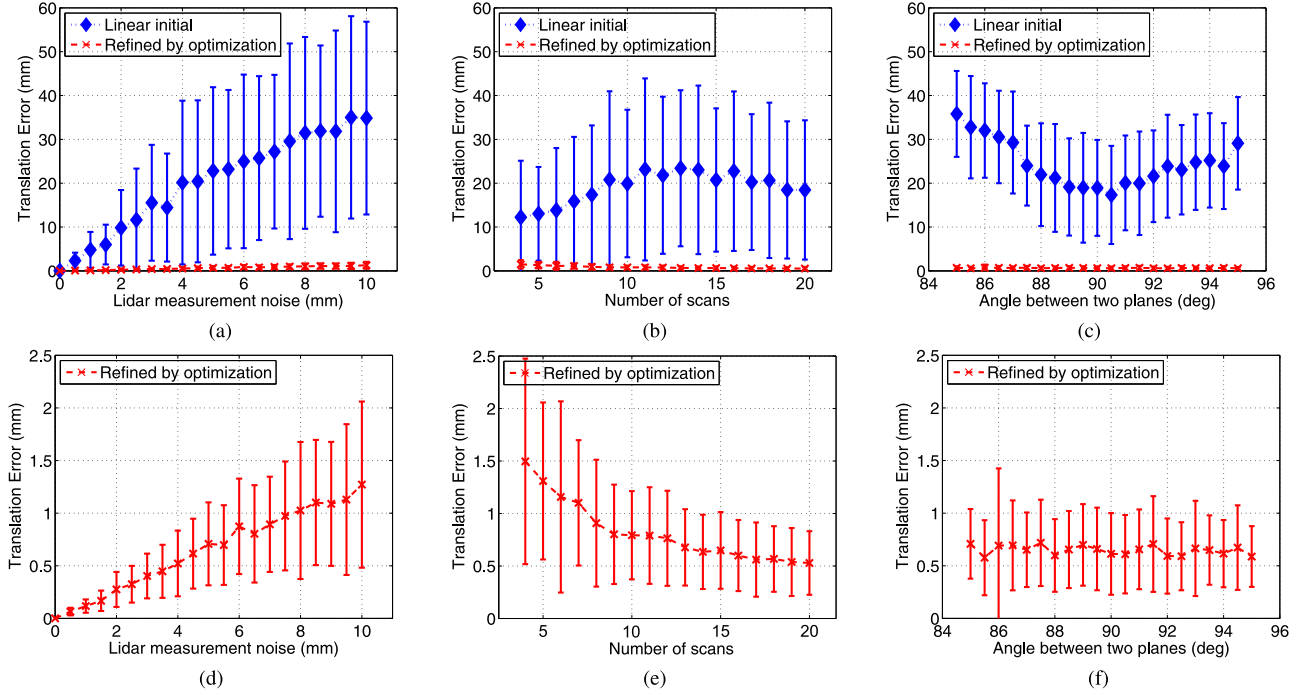


Fig. 7. Performance analysis using translation errors. The results are analyzed in the same ways as those shown in Fig. 6.

after the addition of the Gaussian noise of $N(0, 5)(\text{mm})$ to the range data. A set of four scans is the minimum number of scans for the proposed method because each scan gives two different poses from the two planes. The angle between the two planes is changed from 85° to 95° , as shown in Fig. 6(c), to demonstrate the performance of the proposed algorithm when the assumption of orthogonality is broken. Ten scans are used for the measurement and angle noise experiments.

In the experiment, the initial and refined values of the extrinsic parameters computed using the proposed algorithm are compared with the ground truth. The difference in rotation is measured according to the angle of the difference $\delta \mathbf{R}$ between the true rotation $\mathbf{R}_{\text{true}}$ and the measured rotation $\mathbf{R}_{\text{measured}}$ as $\delta \mathbf{R} = \mathbf{R}_{\text{true}} * \mathbf{R}_{\text{measured}}^{-1}$. The angle of $\delta \mathbf{R}$ expressed in the angle-axis representation and the L_2 norm of the translation are computed as the rotation and translation errors, respectively.

In Fig. 6, we performed 1000 trials and computed the mean and standard deviation of the rotation errors. Blue and red lines indicate the performance of the initial and refined results, respectively. As shown in Fig. 6(a), the rotation error of the initial results increases smoothly as the Gaussian noise increases. Fig. 6(b) shows that changing the number of scans does not have much effect on the accuracy of the initial results. The red lines show that the proposed optimization method reduces the rotation error significantly in all cases. Fig. 6(c) shows that the proposed algorithm is very sensitive to errors in the orthogonality of the two planes. The nonlinear optimization resolves this sensitivity issue because it optimizes the angle between the planes as well. Fig. 6(d)–(f) shows the refined results in detail. Translation errors are also analyzed with the same conditions. As shown in Fig. 7, the translation estimation has a tendency similar to that of the rotation.

We performed another experiment to determine efficient capture motions for the proposed method. In Fig. 8(a), we generated one of three rotation angles in a wider interval than that of the others. For example, “ $R_x 30 R_y 10 R_z 10$ ” means that the rotation angle on the x -axis of the L_1 -to-world transformation is generated in $(-30, 30)(\text{degrees})$, while those angles on the y - and z -axes are generated in $(-10, 10)(\text{degrees})$. The translation parameters are generated at a fixed interval $(-0.5, 0.5)(\text{m})$. The world coordinate system is defined by the three orthogonal vectors of $\mathbf{w}_x$, $\mathbf{w}_y$, and $\mathbf{w}_z$, as shown in Fig. 5.

In Fig. 8(b), we adjusted the intervals in which the translation parameters are generated, similar to Fig. 8(a). For example, “ $T_x 0.7 T_y 0.3 T_z 0.3$ ” means that the translation along the x -axis of the L_1 -to-world transformation is generated in $(-0.7, 0.7)(\text{m})$, while those along the y - and z -axes are generated in $(-0.3, 0.3)(\text{m})$. The rotation angles are generated at a fixed interval $(-20, 20)(\text{degree})$.

The rotation and the translation elements of the relative poses between L_2 -to- L_1 were generated under the same conditions as those used in the previous experiments. We increased the standard deviation of the Gaussian measurement noise of the lidars from 0 to 10(mm) with a step of 0.5(mm) for this experiment. For each noise value, 20 poses were generated for each time, and the distribution of errors computed 1000 times is displayed in Fig. 8. The experiment clearly shows that the rotation on the x - and z -axes of the L_1 -to-world transformation is important for the accuracy of the calibration results.

B. Experiment Using Real Data

For real data experiments, we built a sensor system that consists of two laser sensors (Hokuyo UTM-30LX), L_1 and L_2 , and

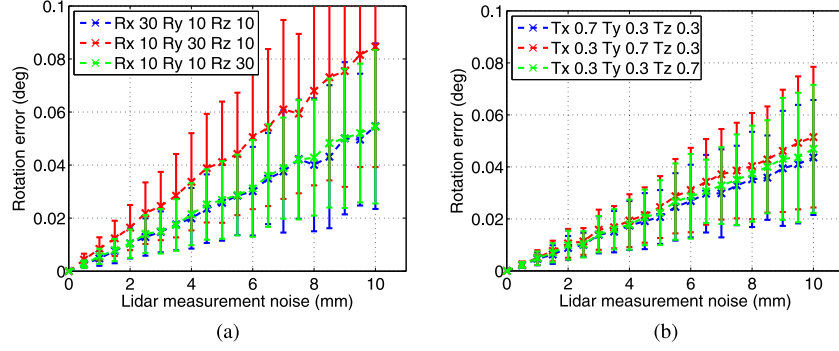


Fig. 8. Experiment in which elements of the sensor-to-world coordinate transformation are important for the accuracy of the final results. Rotations on the x - and z -axes of the L_1 -to-world transformation are important for good results. (a) Lidar measurement noise (mm). (b) Lidar measurement noise (mm).

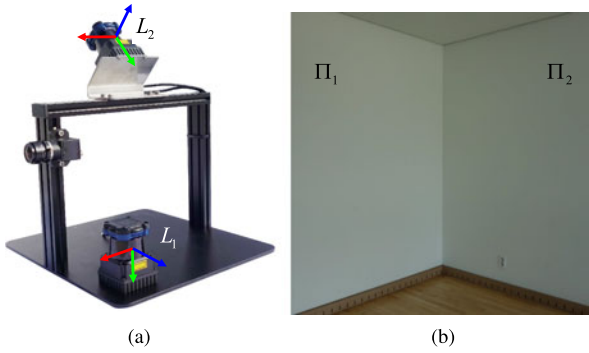


Fig. 9. (a) Sensor configuration: Red, green, and blue arrows denote the x -, y -, and z -axes of each sensor coordinate system, respectively. (b) Experimental environment: We captured the two orthogonal planes (Π_1 and Π_2) for the proposed method.

TABLE I
ESTIMATED EXTRINSIC PARAMETERS USING REAL DATA

	Rotation (rad)
Conventional	0.785485, 0.025077, 0.013777
Proposed	0.787199, 0.019863, 0.009041
	Translation (mm)
Conventional	-98.425873, -276.777585, -172.475253
Proposed	-103.920815, -284.363803, -171.196636

a CCD camera (PointGrey Flea2), as shown in Fig. 9(a). Red, green, and blue arrows represent the x -, y -, and z -axes of each sensor coordinate system, respectively. We scanned two orthogonal planes [see Fig. 9(b)] in various poses including rotations along x - and z -axes.

Using real data, the calibration result of the proposed method is compared with those derived using the conventional method of Zhang and Pless [9], which used an extra camera. In order to compute the relative pose using the conventional method, the relative poses between the two laser sensors and the camera are multiplied; the extra camera serves as a “bridges” between the laser sensors.

Table I shows the results estimated using the proposed method and those determined using the conventional method. The rotation parameters are transformed into a 3×1 vector, as

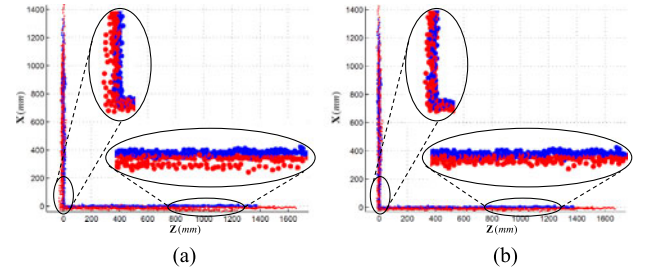


Fig. 10. Comparison experiment between results obtained using the conventional method and those obtained using the proposed method. Scan data for the two laser sensors denoted by blue (L_1) and red (L_2) dots are transformed into the world coordinate system. Since two planes are parallel to the y -axis of the world coordinate system, the scan data must lie on two lines in the xz plane. (a) Conventional. (b) Proposed.

mentioned in Section III-A. The rotation and translation differences between the conventional and the proposed method were estimated as 0.240° and 5.458 mm, respectively. The angle values θ between the two planes for the conventional and the proposed methods were estimated to be 89.869° and 89.871° , respectively.

We transformed the scan data into the world coordinate system using the conventional and the proposed methods. The closeup views shown in Fig. 10 verify that the proposed method is more accurate than the conventional one. We measured the distribution error for both set of results using a plane fitting algorithm. The error was also calculated for both methods [Conventional: 21.235 (mm), Proposed: 19.509 (mm)]. The proposed method used 11 scans to calculate the extrinsic calibration parameters.

In order to evaluate the robustness of the proposed method, we compared results obtained using different numbers of scans (4–10 scans) to results obtained using entire scans (11 scans, see Table I). All possible combinations were selected from a total of eleven scans and used to calculate the results for the proposed method. As shown in Fig. 11, the rotation and translation errors are very small, even if the minimum number of scans is selected. This means that the proposed method always gives a result that is sufficiently close to the global minimum.

Fig. 12 shows the 3-D reconstruction results for a room using the proposed method. The sensor system shown in Fig. 9(a) is

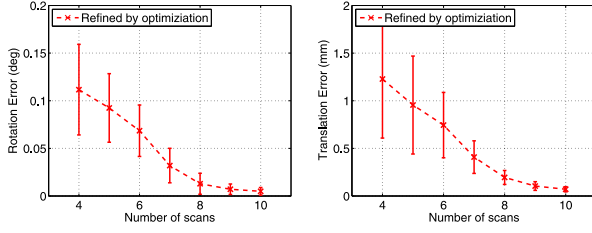


Fig. 11. Errors of results calculated using different numbers of scans, compared with results obtained using entire scans.

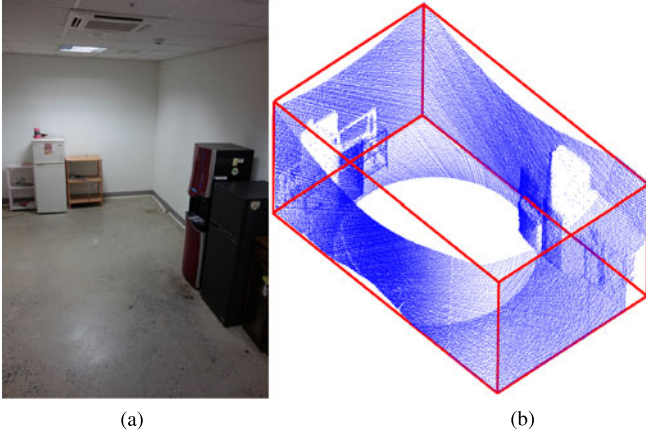


Fig. 12. Three-dimensional reconstruction environment using the calibration results for the proposed method. A real-sized model (i.e., measured by a laser distance meter), denoted by the red box, is registered to the point cloud (blue dots), generated by accumulating laser scans of L_2 . (a) Reconstruction space. (b) Three-dimensional model.

attached to a mobile platform and L_1 scans horizontally for the motion estimation. We employed the iterative closest point [22] algorithm for the registration of the scan data. The scan data of L_2 are accumulated using the calculated motion and calibration results. The 3-D point cloud data for the room were captured five times at different positions. We measured three dimensions of the reconstructed room and calculated the mean value of each dimension. We also measured the three dimensions of the room, as shown in Fig. 12(a), using a laser distance meter (Bosch GLM 80). Each dimension was also measured five times at different positions, and the mean value of these measurements is referred to as the ground truth. As shown in Fig. 12(b), one of the 3-D models determined using the proposed method is well fitted to the measured red box. The size values of the room, as determined using a laser distance meter, and the conventional and proposed methods are shown in Table II. The value inside brackets is the standard deviation of the five trials. The error of the proposed method is smaller than that of the conventional method.

IV. DEGENERATE CASES

In this section, we present degenerate cases for the proposed method. We show that the measurement matrix in (11) can be decomposed into two submatrices: one from the plane poses, and the other from the relative transformation between the two lidars.

TABLE II
RESULTS OF THE ROOM MEASUREMENT (UNIT : mm)

	Width (std.)	Length (std.)	Height (std.)
Laser distance meter	2973.1 (1.92)	4918.7 (2.47)	2374.7 (6.73)
Conventional	2965.1 (2.16)	4908.4 (2.73)	2391.9 (11.27)
Proposed	2968.4 (2.04)	4910.2 (2.70)	2377.8 (10.64)

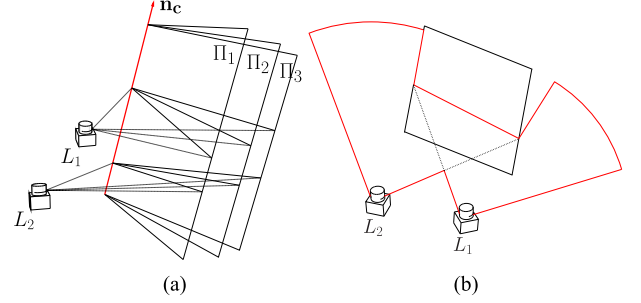


Fig. 13. Degenerate cases of the proposed method. (a) Plane poses: if there exists a common vector $\mathbf{n}_c$ that is orthogonal to the plane normal vector $\mathbf{n}_i$ for every scanned planes. (b) Relative pose: if the two scan planes of L_1 and L_2 are identical.

By analyzing each submatrix, we investigate two degenerate cases of capture motion and relative pose as shown in Fig. 13.

A. Decomposition of the Measurement Matrix

A measurement vector $\mathbf{A}_i$ in (11) is expressed simply:

$$\mathbf{A}_i = \begin{bmatrix} \alpha_1 \alpha_6 & \alpha_2 \alpha_6 & \alpha_3 \alpha_5 & \alpha_4 \alpha_5 & \alpha_1 \alpha_3 & \alpha_1 \alpha_4 & \alpha_2 \alpha_3 & \alpha_2 \alpha_4 \end{bmatrix} \quad (33)$$

where we defined in (7) that

$$\begin{aligned} \alpha_1 &\equiv \Delta x_{21} = \mathbf{e}_1^\top \Delta \mathbf{q}_{21} \\ \alpha_2 &\equiv \Delta z_{21} = \mathbf{e}_3^\top \Delta \mathbf{q}_{21} \\ \alpha_3 &\equiv \Delta x'_{43} = \mathbf{e}_1^\top \mathbf{R}^\top \Delta \mathbf{q}_{43} \\ \alpha_4 &\equiv \Delta z'_{43} = \mathbf{e}_3^\top \mathbf{R}^\top \Delta \mathbf{q}_{43} \\ \alpha_5 &\equiv (z_1 x_2 - x_1 z_2) = \mathbf{e}_2^\top [\mathbf{q}_1]_\times \mathbf{q}_2 \\ \alpha_6 &\equiv (z'_3 x'_4 - x'_3 z'_4) = \mathbf{e}_2^\top [\mathbf{q}'_3]_\times \mathbf{q}'_4. \end{aligned} \quad (34)$$

The unit vectors $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_3$ are in x -, y -, and z -direction on L_1 coordinate, respectively. We interpret each component in the measurement vector in a geometric way.

A scene plane Π_i is represented as $(\mathbf{n}_i, d_i)$ in L_1 coordinate system, i.e., $\mathbf{n}_i$ is the unit normal vector of the plane Π_i and d_i is the distance from the origin of L_1 . In the rest of this paper, i means the i th pose captured by L_1 and L_2 . Since we have assumed that the scanning plane of a laser sensor is equal to its xz plane, the normal vectors of the scan planes of L_1 and L_2 are $\mathbf{e}_2$ and $\mathbf{r}_2$, respectively, where $\mathbf{r}_2$ is the second column of the rotation matrix $\mathbf{R}$.

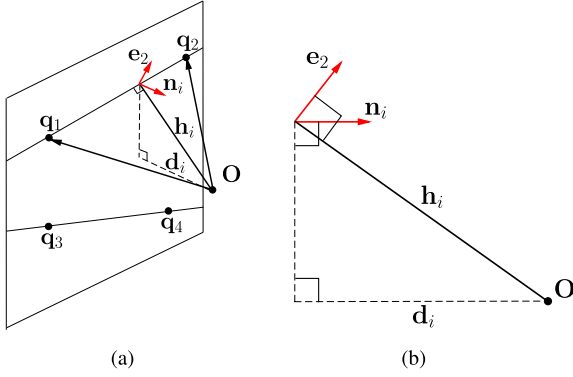


Fig. 14. Relation between the cross product $[\mathbf{q}_1] \times \mathbf{q}_2$ and the distance d_i to the plane from the origin of L_1 .

A direction vector of the line should be orthogonal to both of the lidar scan plane and the scene plane, and thus

$$\begin{aligned} \Delta \mathbf{q}_{21} &= |\Delta \mathbf{q}_{21}| \frac{[\mathbf{e}_2] \times \mathbf{n}_i}{\|[\mathbf{e}_2] \times \mathbf{n}_i\|} = s_1 [\mathbf{e}_2] \times \mathbf{n}_i \\ \Delta \mathbf{q}_{43} &= |\Delta \mathbf{q}_{43}| \frac{[\mathbf{r}_2] \times \mathbf{n}_i}{\|[\mathbf{r}_2] \times \mathbf{n}_i\|} = s_2 [\mathbf{r}_2] \times \mathbf{n}_i. \end{aligned} \quad (35)$$

The common constant values s_1 and s_2 are defined that

$$s_1 \equiv \frac{|\Delta \mathbf{q}_{21}|}{\|[\mathbf{e}_2] \times \mathbf{n}_i\|}, \quad s_2 \equiv \frac{|\Delta \mathbf{q}_{43}|}{\|[\mathbf{r}_2] \times \mathbf{n}_i\|}. \quad (36)$$

Thus, the components α_1 – α_4 can be simplified as

$$\begin{aligned} \alpha_1 &= \mathbf{e}_1^\top \Delta \mathbf{q}_{21} = s_1 \mathbf{e}_1^\top [\mathbf{e}_2] \times \mathbf{n}_i = s_1 \mathbf{e}_3^\top \mathbf{n}_i \\ \alpha_2 &= \mathbf{e}_3^\top \Delta \mathbf{q}_{21} = s_2 \mathbf{e}_3^\top [\mathbf{e}_2] \times \mathbf{n}_i = -s_1 \mathbf{e}_1^\top \mathbf{n}_i \\ \alpha_3 &= \mathbf{e}_1^\top \mathbf{R}^\top \Delta \mathbf{q}_{43} = s_2 \mathbf{e}_1^\top \mathbf{R}^\top [\mathbf{r}_2] \times \mathbf{n}_i = s_2 \mathbf{r}_3^\top \mathbf{n}_i \\ \alpha_4 &= \mathbf{e}_3^\top \mathbf{R}^\top \Delta \mathbf{q}_{43} = s_2 \mathbf{e}_3^\top \mathbf{R}^\top [\mathbf{r}_2] \times \mathbf{n}_i = -s_2 \mathbf{r}_1^\top \mathbf{n}_i. \end{aligned} \quad (37)$$

The value of α_5 in (34) is calculated as follows:

$$\alpha_5 = \mathbf{e}_2^\top [\mathbf{q}_1] \times \mathbf{q}_2 = \|\mathbf{q}_1 \times \mathbf{q}_2\| \quad (38)$$

because the cross product $[\mathbf{q}_1] \times \mathbf{q}_2$ is parallel to the unit vector $\mathbf{e}_2$. The magnitude of the cross product is equal to the area of the parallelogram whose sides are $\mathbf{q}_1$ and $\mathbf{q}_2$. The area can also be computed from the area of the triangle Oq_1q_2 by

$$\alpha_5 = \|\mathbf{q}_1 \times \mathbf{q}_2\| = |\Delta \mathbf{q}_{21}| h_i \quad (39)$$

where h_i is the orthogonal distance from the origin (center of L_1) to the line containing $\Delta \mathbf{q}_{21}$.

Fig. 14 shows that the distance d_i from the origin to the plane Π_i is related to h_i as

$$d_i \equiv \|[\mathbf{e}_2] \times \mathbf{n}_i\| h_i \quad (40)$$

and it gives

$$\begin{aligned} \alpha_5 &= \|\mathbf{q}_1 \times \mathbf{q}_2\| = |\Delta \mathbf{q}_{21}| h_i \\ &= s_1 \|[\mathbf{e}_2] \times \mathbf{n}_i\| h_i \\ &= s_1 d_i. \end{aligned} \quad (41)$$

Similarly, α_6 can be derived as

$$\alpha_6 = \mathbf{e}_2^\top [\mathbf{q}_3'] \times \mathbf{q}_4' = -s_2 d_i' \quad (42)$$

where the distance d_i' is from the origin of L_2 to the plane Π_i , which is related as

$$d_i' = \mathbf{n}_i^\top \mathbf{t} + d_i. \quad (43)$$

Finally, the measurement vector $\mathbf{A}_i$ in (33) can be expressed as

$$\mathbf{A}_i = \begin{bmatrix} \alpha_1 \alpha_6 \\ \alpha_2 \alpha_6 \\ \alpha_3 \alpha_5 \\ \alpha_4 \alpha_5 \\ \alpha_1 \alpha_3 \\ \alpha_1 \alpha_4 \\ \alpha_2 \alpha_3 \\ \alpha_2 \alpha_4 \end{bmatrix}^\top = s_1 s_2 \begin{bmatrix} -\mathbf{e}_3^\top \mathbf{n}_i \mathbf{n}_i^\top \mathbf{t} - d_i \mathbf{n}_i^\top \mathbf{e}_3 \\ \mathbf{e}_1^\top \mathbf{n}_i \mathbf{n}_i^\top \mathbf{t} + d_i \mathbf{n}_i^\top \mathbf{e}_1 \\ -d_i \mathbf{n}_i^\top \mathbf{r}_3 \\ d_i \mathbf{n}_i^\top \mathbf{r}_1 \\ \mathbf{e}_3^\top \mathbf{n}_i \mathbf{n}_i^\top \mathbf{r}_3 \\ -\mathbf{e}_3^\top \mathbf{n}_i \mathbf{n}_i^\top \mathbf{r}_1 \\ -\mathbf{e}_1^\top \mathbf{n}_i \mathbf{n}_i^\top \mathbf{r}_3 \\ \mathbf{e}_1^\top \mathbf{n}_i \mathbf{n}_i^\top \mathbf{r}_1 \end{bmatrix}^\top. \quad (44)$$

Now, we take a look into the right-hand side vector in (44). Because the scale factors s_1 and s_2 can be disregarded, it has mainly two components: plane parameters $\mathbf{n}_i$ and d_i , and the relative pose $\mathbf{R}$ and $\mathbf{t}$. The plane normal $\mathbf{n}_i = (a_i, b_i, c_i)$ and the distance d_i appear in the measurement vector only in eight combinations of

$$\mathbf{m}_i = [a_i^2 \quad a_i b_i \quad a_i c_i \quad b_i c_i \quad c_i^2 \quad a_i d_i \quad b_i d_i \quad c_i d_i]. \quad (45)$$

Then, the measurement vector $\mathbf{A}_i$ can be expressed in general as

$$\mathbf{A}_i = \mathbf{m}_i \mathbf{G}_{8 \times 8} \quad (46)$$

where

$$\mathbf{G}_{8 \times 8} = \begin{bmatrix} 0 & t_x & 0 & 0 & 0 & 0 & -r_{13} & r_{11} \\ 0 & t_y & 0 & 0 & 0 & 0 & -r_{23} & r_{21} \\ -t_x & t_z & 0 & 0 & r_{13} & -r_{11} & -r_{33} & r_{31} \\ -t_y & 0 & 0 & 0 & r_{23} & -r_{21} & 0 & 0 \\ -t_z & 0 & 0 & 0 & r_{33} & -r_{31} & 0 & 0 \\ 0 & 1 & -r_{13} & r_{11} & 0 & 0 & 0 & 0 \\ 0 & 0 & -r_{23} & r_{21} & 0 & 0 & 0 & 0 \\ -1 & 0 & -r_{33} & r_{31} & 0 & 0 & 0 & 0 \end{bmatrix} \quad (47)$$

which is referred to as a *generator*. This generator is only dependent to the relative poses. In fact, the desired solution

$$\begin{bmatrix} r_{12} & r_{32} & r_{21} & r_{23} & \dots \\ r_{31} t_y - r_{21} t_z & r_{33} t_y - r_{23} t_z & r_{21} t_x - r_{11} t_y & r_{23} t_x - r_{13} t_y \end{bmatrix} \quad (48)$$

is the right null vector of the generator $\mathbf{G}_{8 \times 8}$, and the rank of $\mathbf{G}_{8 \times 8}$ is seven at most.

The measurement matrix $\mathbf{A}$ stacking all the measurement vectors $\mathbf{A}_i$ can be written as

$$\mathbf{A} = \mathbf{M}_{n \times 8} \mathbf{G}_{8 \times 8} \quad (49)$$

where $\mathbf{M}_{n \times 8}$ is the stacked coefficient vectors $\mathbf{m}_i$ for each plane Π_i .

From (49), we show that the $\mathbf{A}\mathbf{x} = \mathbf{0}$ system does not yield a unique solution if $\text{rank}(\mathbf{M}_{n \times 8}) < 7$ or $\text{rank}(\mathbf{G}_{8 \times 8}) < 7$. We will investigate two important cases using the decomposition.

B. Critical Plane Poses

The first case is about the plane poses with respect to the sensor coordinate system, which is related to the matrix $\mathbf{M}_{n \times 8}$.

Lemma IV.1: If there exists a vector $\mathbf{n}_c$ such that $\mathbf{n}_c^\top \mathbf{n}_i = 0$ for all i , $\text{rank}(\mathbf{A}) \leq 5$.

Proof: If there exists a common orthogonal vector $\mathbf{n}_c$ to the plane normal vector $\mathbf{n}_i$, any unit normal vector $\mathbf{n}_i$ is expressed with a parameter t as

$$\mathbf{n}_i = \mathbf{u} \cos t + \mathbf{v} \sin t \quad (50)$$

where $\mathbf{u}$ and $\mathbf{v}$ are orthogonal basis vectors on the plane, i.e., $\mathbf{n}_c^\top \mathbf{u} = \mathbf{n}_c^\top \mathbf{v} = 0$.

$$\begin{aligned} \mathbf{n}_i \mathbf{n}_i^\top &= (\mathbf{u} \cos t + \mathbf{v} \sin t)(\mathbf{u} \cos t + \mathbf{v} \sin t)^\top \\ &= \cos^2(t) \mathbf{u} \mathbf{u}^\top + \sin^2(t) \mathbf{v} \mathbf{v}^\top \\ &\quad + \cos(t) \sin(t) (\mathbf{u} \mathbf{v}^\top + \mathbf{v} \mathbf{u}^\top). \end{aligned} \quad (51)$$

A row vector in $\mathbf{M}_{n \times 8}$ is expressed as

$$\begin{bmatrix} a_i^2 \\ a_i b_i \\ a_i c_i \\ b_i c_i \\ c_i^2 \\ a_i d_i \\ b_i d_i \\ c_i d_i \end{bmatrix}^\top = \begin{bmatrix} d_i \cos t \\ d_i \sin t \\ \cos^2 t \\ \sin^2 t \\ \cos t \sin t \end{bmatrix}^\top$$

$$\begin{bmatrix} 0 & 0 & \mathbf{e}_1^\top \mathbf{u} \mathbf{u}^\top \mathbf{e}_1 & \mathbf{e}_1^\top \mathbf{v} \mathbf{v}^\top \mathbf{e}_1 & \mathbf{e}_1^\top (\mathbf{u} \mathbf{v}^\top + \mathbf{v} \mathbf{u}^\top) \mathbf{e}_1 \\ 0 & 0 & \mathbf{e}_1^\top \mathbf{u} \mathbf{u}^\top \mathbf{e}_2 & \mathbf{e}_1^\top \mathbf{v} \mathbf{v}^\top \mathbf{e}_2 & \mathbf{e}_1^\top (\mathbf{u} \mathbf{v}^\top + \mathbf{v} \mathbf{u}^\top) \mathbf{e}_2 \\ 0 & 0 & \mathbf{e}_1^\top \mathbf{u} \mathbf{u}^\top \mathbf{e}_3 & \mathbf{e}_1^\top \mathbf{v} \mathbf{v}^\top \mathbf{e}_3 & \mathbf{e}_1^\top (\mathbf{u} \mathbf{v}^\top + \mathbf{v} \mathbf{u}^\top) \mathbf{e}_3 \\ 0 & 0 & \mathbf{e}_2^\top \mathbf{u} \mathbf{u}^\top \mathbf{e}_3 & \mathbf{e}_2^\top \mathbf{v} \mathbf{v}^\top \mathbf{e}_3 & \mathbf{e}_2^\top (\mathbf{u} \mathbf{v}^\top + \mathbf{v} \mathbf{u}^\top) \mathbf{e}_3 \\ 0 & 0 & \mathbf{e}_3^\top \mathbf{u} \mathbf{u}^\top \mathbf{e}_3 & \mathbf{e}_3^\top \mathbf{v} \mathbf{v}^\top \mathbf{e}_3 & \mathbf{e}_3^\top (\mathbf{u} \mathbf{v}^\top + \mathbf{v} \mathbf{u}^\top) \mathbf{e}_3 \\ \mathbf{e}_1^\top \mathbf{u} & \mathbf{e}_1^\top \mathbf{v} & 0 & 0 & 0 \\ \mathbf{e}_2^\top \mathbf{u} & \mathbf{e}_2^\top \mathbf{v} & 0 & 0 & 0 \\ \mathbf{e}_3^\top \mathbf{u} & \mathbf{e}_3^\top \mathbf{v} & 0 & 0 & 0 \end{bmatrix}^\top. \quad (52)$$

In this case, any row vector in $\mathbf{M}_{n \times 8}$ is determined by the five-coefficient vector, and obviously, $\text{rank}(\mathbf{M}_{n \times 8}) \leq 5$. Thus, the maximum rank of $\mathbf{A}$ is 5. $\square$

This case happens when a scene plane rotates on a single axis, while the laser scanners are fixed as shown in Fig. 13(a).

C. Critical Relative Pose

The other case is about the matrix $\mathbf{G}_{8 \times 8}$, which is related to the relative poses between lidar sensors L_1 and L_2 .

Lemma IV.2: $\text{rank}(\mathbf{G}_{8 \times 8}) < 7$ if and only if the two scan planes of L_1 and L_2 are identical (i.e., $r_{22} = 1$ and $t_y = 0$), and $\text{rank}(\mathbf{G}_{8 \times 8}) \leq 5$.

Proof: The generator $\mathbf{G}_{8 \times 8}$ can be arranged as

$$\begin{bmatrix} 0 & t_y & 0 & 0 & 0 & 0 & -r_{23} & r_{21} \\ -t_y & 0 & 0 & 0 & r_{23} & -r_{21} & 0 & 0 \\ 0 & 0 & -r_{23} & r_{21} & 0 & 0 & 0 & 0 \\ \hline 0 & t_x & 0 & 0 & 0 & 0 & -r_{13} & r_{11} \\ -t_x & t_z & 0 & 0 & r_{13} & -r_{11} & -r_{33} & r_{31} \\ -t_z & 0 & 0 & 0 & r_{33} & -r_{31} & 0 & 0 \\ 0 & 1 & -r_{13} & r_{11} & 0 & 0 & 0 & 0 \\ -1 & 0 & -r_{33} & r_{31} & 0 & 0 & 0 & 0 \end{bmatrix}, \quad (53)$$

which shows the relationship between the rotation parameters r_{21}, r_{23} and the translation t_y in the upper three rows.

One can easily notice that there are 2×2 submatrices of r_{11}, r_{13}, r_{31} and r_{33} noticed by dashed boxes, and they are in the row and sign-flipped form of a 2-D rotation matrix. Thus, the lower five rows cannot be linearly dependent to each other.

If $r_{22} = 1$, which induces

$$r_{21} = r_{12} = r_{23} = r_{32} = 0 \quad (54)$$

and

$$r_{11} = r_{33}, \quad r_{13} = -r_{31}$$

the upper three rows of (53) becomes

$$\begin{bmatrix} 0 & t_y & 0 & 0 & 0 & 0 & 0 & 0 \\ -t_y & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}. \quad (55)$$

These three rows diminish to zero if and only if $t_y = 0$, and this leads to $\text{rank}(\mathbf{G}_{8 \times 8}) \leq 5$.

If $r_{22} \neq 1$, we show that $\text{rank}(\mathbf{G}_{8 \times 8}) = 7$ by proof by contradiction. Assume that $\text{rank}(\mathbf{G}_{8 \times 8})$ has two null space $\mathbf{v}_1$ and $\mathbf{v}_2$ which are not linearly dependent. Let v_{ij} denote the j th element of the vector $\mathbf{v}_i$. From the third row in (53)

$$-r_{23} v_{13} + r_{21} v_{14} = -r_{23} v_{23} + r_{21} v_{24} = 0.$$

Because r_{21} and r_{23} cannot be zero from the condition $r_{22} \neq 1$, we have

$$v_{13} = v_{23} \text{ and } v_{14} = v_{24}. \quad (56)$$

From the seventh and the eighth row in (53) along with (56), we also have

$$v_{11} = v_{21} \text{ and } v_{12} = v_{22}. \quad (57)$$

From (57), the first and second rows in (53) are written as

$$\begin{aligned} v_{12}t_y &= r_{23}v_{17} - r_{21}v_{18} = r_{23}v_{27} - r_{21}v_{28} \\ v_{11}t_y &= r_{23}v_{15} - r_{21}v_{16} = r_{23}v_{25} - r_{21}v_{26} \end{aligned}$$

which give

$$\begin{aligned} r_{21}(v_{28} - v_{18}) - r_{23}(v_{27} - v_{17}) &= 0 \\ r_{21}(v_{26} - v_{16}) - r_{23}(v_{25} - v_{15}) &= 0 \end{aligned}$$

and this concludes that

$$v_{28} - v_{18} = v_{26} - v_{16} \text{ and } v_{27} - v_{17} = v_{25} - v_{15}. \quad (58)$$

The fourth and sixth rows in (53) similarly expands as

$$\begin{aligned} r_{11}(v_{28} - v_{18}) - r_{13}(v_{27} - v_{17}) &= 0 \\ r_{31}(v_{26} - v_{16}) - r_{33}(v_{25} - v_{15}) &= 0 \end{aligned}$$

and by the equality shown in (58), it can be rewritten as

$$\begin{bmatrix} r_{11} & r_{13} \\ r_{21} & r_{23} \\ r_{31} & r_{33} \end{bmatrix} \begin{bmatrix} v_{28} - v_{18} \\ -(v_{27} - v_{17}) \end{bmatrix} = \mathbf{0}. \quad (59)$$

Because the leftmost matrix in (59) has two columns of the rotation matrix, and thus, they cannot have a right null space, it gives

$$\begin{aligned} v_{28} - v_{18} &= 0 \\ v_{27} - v_{17} &= 0. \end{aligned} \quad (60)$$

From (56)–(58) and (60), we conclude that $\mathbf{v}_1 = \mathbf{v}_2$, which contradicts the assumption that $\mathbf{v}_1$ and $\mathbf{v}_2$ are not linearly independent.

Therefore, if $r_{22} \neq 1$, $\text{rank}(\mathbf{G}_{8 \times 8}) = 7$. $\square$

That the critical relative pose of the two scan planes of L_1 and L_2 are identical is shown in Fig. 14(b). It should be noted that there may exist more degenerate cases for the critical plane pose, but there are no more degenerate cases for the critical relative pose. Even in a case in which two scan planes of L_1 and L_2 are parallel (i.e., $r_{22} = 1$ and $t_y \neq 0$), we can still estimate the extrinsic parameters. However, we can say that any further degeneracy can be investigated using the decomposition in (49).

V. CONCLUSION

In this paper, we have presented a novel method for extrinsic calibration of two 2-D lidars. The proposed algorithm uses only range data scanned on two orthogonal planes without any additional sensors or any information about the motion of the laser sensors. The coplanarity of the scan data from the two laser sensors gives a linear equation of their relative pose. The scales of a solution vector obtained via SVD are determined by the orthogonality of the two planes. Two “mirrored” solutions are computed from a proper scale selected using multiple scans. We have demonstrated the accuracy and robustness of the proposed algorithm using synthetic data with noise. The experiment using real data verified that the proposed method provides a solution comparable with that of the conventional method, which uses

an extra camera. We have analyzed two degenerate cases of the proposed method, caused by plane poses and a relative pose.

APPENDIX A

PROPERTIES OF A REDUCED LIDAR ESSENTIAL MATRIX

We have shown in (9) that, if a two lidar scans a single plane, we have a single bilinear constraint as

$$\mathbf{L}_{12}^\top \mathbf{E}_L \mathbf{L}_{34}' = 0. \quad (61)$$

The reduced lidar essential matrix $\mathbf{E}_L$ is defined with parameters of $\mathbf{R}$ and $\mathbf{t}$ as

$$\mathbf{E}_L = \begin{bmatrix} r_{31}t_y - r_{21}t_z & r_{33}t_y - r_{23}t_z & r_{12} \\ r_{21}t_x - r_{11}t_y & r_{23}t_x - r_{13}t_y & r_{32} \\ r_{21} & r_{23} & 0 \end{bmatrix}. \quad (62)$$

We will introduce some properties of the reduced lidar essential matrix and finally show that it is not possible to uniquely determine the pose parameters from the estimated reduced lidar essential matrix.

A. Rank of $\mathbf{E}_L$: We start from its rank condition and its physical meaning.

Lemma VI.1: The rank of the reduced lidar essential matrix $\mathbf{E}_L$ is two at most.

Proof: The i th row and j th column of $\mathbf{E}_L$ are denoted by $\mathbf{E}_L(i, :)$ and $\mathbf{E}_L(:, j)$, respectively. At first, we show that $\mathbf{E}_L(:, 3) \sim r_{23}\mathbf{E}_L(:, 1) - r_{21}\mathbf{E}_L(:, 2)$.

$$\begin{aligned} & r_{23}\mathbf{E}_L(:, 1) - r_{21}\mathbf{E}_L(:, 2) \\ &= \begin{bmatrix} r_{23}(r_{31}t_y - r_{21}t_z) - r_{21}(r_{33}t_y - r_{23}t_z) \\ r_{23}(r_{21}t_x - r_{11}t_y) - r_{21}(r_{23}t_x - r_{13}t_y) \\ r_{21}r_{23} - r_{21}r_{23} \end{bmatrix} \\ &= \begin{bmatrix} r_{23}r_{31} - r_{21}r_{33} \\ r_{23}r_{13} - r_{21}r_{11} \\ 0 \end{bmatrix} t_y = \begin{bmatrix} C_{12} \\ C_{32} \\ 0 \end{bmatrix} t_y. \end{aligned} \quad (63)$$

Due to Cramer’s rule, the elements of a rotation matrix $r_{12} = C_{12}$ and $r_{32} = C_{32}$, where C_{ij} is the cofactor of the rotation matrix at (i, j) . Thus, the vector is the same with $\mathbf{E}_L(:, 1)$ up to scale, if $t_y \neq 0$.

For the row vector, similarly we can show that $\mathbf{E}_L(3, :) \sim r_{32}\mathbf{E}_L(1, :) - r_{12}\mathbf{E}_L(2, :)$ by

$$\begin{aligned} & r_{32}\mathbf{E}_L(1, :) - r_{12}\mathbf{E}_L(2, :) \\ &= \begin{bmatrix} r_{32}(r_{31}t_y - r_{21}t_z) - r_{12}(r_{21}t_x - r_{11}t_y) \\ r_{32}(r_{33}t_y - r_{23}t_z) - r_{12}(r_{23}t_x - r_{13}t_y) \\ r_{12}r_{32} - r_{12}r_{32} \end{bmatrix}^\top \\ &= \begin{bmatrix} -r_{21}r_{12}t_x + (r_{12}r_{11} + r_{32}r_{31})t_y - r_{21}r_{32}t_z \\ -r_{23}r_{12}t_x + (r_{12}r_{13} + r_{32}r_{33})t_y - r_{23}r_{32}t_z \\ 0 \end{bmatrix}^\top. \end{aligned} \quad (64)$$

From the orthonormality of a rotation matrix, $\mathbf{r}_1^\top \mathbf{r}_2 = \mathbf{r}_1^\top \mathbf{r}_3 = 0$, $(r_{12}r_{11} + r_{32}r_{31}) = -r_{21}r_{22}$, and $(r_{12}r_{13} + r_{32}r_{33}) = -r_{23}r_{22}$. Thus, the combination of the row vector can be written as

$$\begin{bmatrix} r_{32}(r_{31}t_y - r_{21}t_z) - r_{12}(r_{21}t_x - r_{11}t_y) \\ r_{32}(r_{33}t_y - r_{23}t_z) - r_{12}(r_{23}t_x - r_{13}t_y) \\ r_{12}r_{32} - r_{12}r_{32} \end{bmatrix}^\top = \begin{bmatrix} r_{21} \\ r_{23} \\ 0 \end{bmatrix}^\top \mathbf{r}_2^\top \mathbf{t}, \quad (65)$$

if $\mathbf{r}_2^\top \mathbf{t} \neq 0$.

The third row and column are linearly dependent to the first and second rows and columns, respectively. It proves the rank of the matrix should be two at most. $\square$

This lemma can be interpreted as follows.

Corollary VI.2: From the estimated reduced lidar essential matrix $\mathbf{E}_L$ defined up to scale, we can estimate t_y/λ and $\mathbf{r}_2^\top \mathbf{t}/\lambda$ from the row and column combination of $\mathbf{E}_L$, respectively, where λ is the proper scale factor.

B. Linear Constraints on the Pose Parameters: From (11), the linear solution $\mathbf{v}$ is defined up to scale. In this section, we obtain some linear constraints between the estimated $\mathbf{v}$ and the pose parameters by exposing rotation constraints.

Dividing by λ on both sides, the orthogonality $\mathbf{r}_1^\top \mathbf{r}_2 = \mathbf{r}_2^\top \mathbf{r}_3 = 0$ in columns and in rows gives us four linear constraints as

$$\begin{aligned} r_{11}v_1 + r_{22}v_3 + r_{31}v_4 &= 0 \\ r_{13}v_1 + r_{33}v_2 + r_{22}v_4 &= 0 \\ r_{22}v_1 + r_{11}v_3 + r_{13}v_4 &= 0 \\ r_{22}v_2 + r_{31}v_3 + r_{33}v_4 &= 0. \end{aligned} \quad (66)$$

The rotation component r_{12} can be expressed with a cofactor as

$$r_{12} = C_{12} = -r_{21}r_{33} + r_{23}r_{31} = \lambda v_1. \quad (67)$$

Dividing both sides with λ ,

$$v_1 = -r_{33}v_3 + r_{31}v_4 \quad (68)$$

which is a linear constraint between the solution vector $\mathbf{v}$ and some elements of the rotation matrix. Similarly, we can derive the linear relations for r_{32} , r_{21} , and r_{23} . From the cofactor relation, we have other four constraints.

On the other hand, Corollary VI.2 provides $E_1 \equiv t_y/\lambda$ and

$$\begin{aligned} E_2 \equiv \mathbf{r}_2^\top \mathbf{t}/\lambda &= t_x r_{12}/\lambda + r_{22}t_y/\lambda + t_z r_{32}/\lambda \\ &= t_x v_1 + r_{22}E_1 + t_z v_2 \end{aligned} \quad (69)$$

which is another linear constraint. Note that E_1 and E_2 can be estimated from the estimated reduced lidar essential matrix.

From v_5 to v_8 , we can get

$$\begin{aligned} v_5 &= r_{31}t_y/\lambda - t_z r_{21}/\lambda = r_{31}E_1 - t_z v_3 \\ v_6 &= r_{33}t_y/\lambda - t_z r_{23}/\lambda = r_{33}E_1 - t_z v_4 \\ v_7 &= t_x r_{21}/\lambda - r_{11}t_y/\lambda = t_x v_3 - r_{11}E_1 \\ v_8 &= t_x r_{23}/\lambda - r_{13}t_y/\lambda = t_x v_4 - r_{13}E_1 \end{aligned} \quad (70)$$

which are four linear relations between the scaled matrix elements and the remaining unknowns.

By stacking all the 13 relations, the unknowns $\mathbf{p} = [r_{22}, r_{11}, r_{13}, r_{31}, r_{33}, t_x, t_z]^\top$ are linearly constrained as

$$\mathbf{A}_m \mathbf{p} = \mathbf{b}_m \quad (71)$$

where

$$\mathbf{A}_m = \begin{bmatrix} v_1 & v_3 & v_4 & & & & & & \\ v_2 & & & v_3 & v_4 & & & & \\ v_3 & v_1 & & v_2 & & & & & \\ v_4 & & v_1 & v_2 & & & & & \\ & & & v_4 & -v_3 & & & & \\ & -v_4 & v_3 & & & & & & \\ & & v_2 & & -v_1 & & & & \\ & -v_2 & & v_1 & & & & & \\ & & & E_1 & & & & & \\ & & & & E_1 & & & -v_3 & \\ & -E_1 & & & & & v_3 & & -v_4 \\ & & -E_1 & & & & v_4 & & \\ E_1 & & & & & & v_1 & v_2 & \end{bmatrix}, \quad \mathbf{b}_m = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \\ v_7 \\ v_8 \\ E_2 \end{bmatrix}. \quad (72)$$

Although $\mathbf{A}_m$ is 13×7 , we show that the unknown pose parameters are still underconstrained by the linear form (72).

Lemma VI.3: The rank of the matrix $\mathbf{A}_m$ in (72) is six at most.

Proof: We show that there exists a null space of $\mathbf{A}_m$ to show the rank-deficiency. We start from the upper 8×5 part $\mathbf{A}_{m8}$ of the matrix $\mathbf{A}_m$ and the vector $\mathbf{b}_m$, which are related to the five rotation parameters.

We can simply show that there are two possible solutions. The first solution $\mathbf{p}_1 = [r_{22}, r_{11}, r_{13}, r_{31}, r_{33}]^\top$ is obtained trivially. By investigating the cofactor relation of the rotation matrix, we can find that there exists another solution

$$\mathbf{p}_2 = \frac{1}{r_{22}} [1, r_{33}, -r_{31}, -r_{13}, r_{11}]^\top \quad (73)$$

satisfying the upper eight linear equations.

From the two solutions, the null space $\mathbf{N}_{A8}$ of $\mathbf{A}_{m8}$ is given as

$$\mathbf{N}_{A8} = \begin{bmatrix} 1 - r_{22}^2 \\ r_{33} - r_{11}r_{22} \\ -r_{31} - r_{13}r_{22} \\ -r_{13} - r_{31}r_{22} \\ r_{11} - r_{33}r_{22} \end{bmatrix} = \begin{bmatrix} 1 - r_{22}^2 \\ -r_{12}r_{21} \\ -r_{12}r_{23} \\ -r_{21}r_{32} \\ -r_{23}r_{32} \end{bmatrix}^\top \quad (74)$$

which concludes that the 8×5 matrix $\mathbf{A}_{m8}$ has rank 4 at most.

The second part of the proof shows that there exists a null space of the lower five rows derived from the $\mathbf{N}_{A8}$. One can notice that the matrix $\mathbf{A}_m$ can be expressed as a block matrix:

$$\mathbf{A}_m = \begin{bmatrix} \mathbf{A}_{m8} & \mathbf{0} \\ \mathbf{A}_{m5} & \mathbf{V}_{5 \times 2} \end{bmatrix} \quad (75)$$

where $\mathbf{A}_{m5}$ is the lower left 5×5 part of $\mathbf{A}_m$. If there exists a two-vector $\mathbf{b}_m$ satisfying $\mathbf{A}_{m5}\mathbf{N}_{A8} + \mathbf{V}_{5 \times 2}\mathbf{b}_m = \mathbf{0}$, the maximum rank of $\mathbf{A}_m$ should be six. A vector $\mathbf{b}_m = [-r_{12}t_y \ -r_{32}t_y]^\top$ satisfies the condition, and the null space of $\mathbf{A}_m$ in the analytic form is given as

$$\text{Null}(\mathbf{A}_m) = \begin{bmatrix} 1 - r_{22}^2 \\ -r_{12}r_{21} \\ -r_{12}r_{23} \\ -r_{21}r_{32} \\ -r_{23}r_{32} \\ -r_{12}t_y \\ -r_{32}t_y \end{bmatrix} \quad (76)$$

which ends the proof. $\square$

C. Impossibility of Unique Determination of Pose Parameters From a Reduced Lidar Essential Matrix: In this section, we show that the pose parameters are not uniquely determined from a reduced lidar essential matrix. We have shown that some pose parameters are linearly constrained by the estimated reduced lidar essential matrix defined up to scale, but have also shown that there exists a one-dimensional ambiguity. However, we have not yet fully used the orthonormality of the rotation matrix for estimating the pose parameters from the lidar essential matrix.

Let us call $\mathbf{N}_{Am}$ the null space of $\mathbf{A}_m$ and $\mathbf{X}_m \equiv \mathbf{A}_m^\dagger \mathbf{b}_m$, where $\mathbf{A}_m^\dagger$ is a pseudoinverse of $\mathbf{A}_m$. Then, the rotation matrix $\mathbf{R}$ is expressed as

$$\mathbf{R} = \lambda \begin{bmatrix} 0 & v_1 & 0 \\ v_3 & 0 & v_4 \\ 0 & v_2 & 0 \end{bmatrix} + \alpha \begin{bmatrix} n_2 & 0 & n_3 \\ 0 & n_1 & 0 \\ n_4 & 0 & n_5 \end{bmatrix} + \begin{bmatrix} x_2 & 0 & x_3 \\ 0 & x_1 & 0 \\ x_4 & 0 & x_5 \end{bmatrix} \quad (77)$$

where n_i and x_i denotes the elements of $\mathbf{N}_{Am}$ and $\mathbf{X}_m$, respectively. From $\mathbf{R}\mathbf{R}^\top = \mathbf{I}$, we have the following four equations at (2, 2), (1, 1), (3, 3), and (1, 3), which are not used yet:

$$\begin{aligned} n_1^2 \alpha^2 + (2n_1x_1)\alpha + (v_3^2 + v_4^2)\lambda^2 + x_1^2 - 1 &= 0 \\ (n_2^2 + n_3^2)\alpha^2 + (2n_2x_2 + 2n_3x_3)\alpha + v_1^2\lambda^2 + x_2^2 + x_3^2 - 1 &= 0 \\ (n_4^2 + n_5^2)\alpha^2 + (2n_4x_4 + 2n_5x_5)\alpha + v_2^2\lambda^2 + x_4^2 + x_5^2 - 1 &= 0 \\ (n_2n_4 + n_3n_5)\alpha^2 + (n_2x_4 + n_4x_2 + n_3x_5 + n_5x_3)\alpha \\ + v_1v_2\lambda^2 + x_2x_4 + x_3x_5 &= 0. \end{aligned} \quad (78)$$

Note that the other orthogonality properties between vectors in the rotation matrix have been exposed in (72).

Each equation in (78) expresses a conic for the unknown α and the scale factor λ . We have four conics with two unknowns, and thus, it seems to be solvable at the first sight. However, all the four curves are identical.

The curves in (78) can be written as

$$\mathbf{C}_{co} [\alpha^2 \ \alpha \ \lambda^2 \ 1]^\top = \mathbf{0} \quad (79)$$

where (80), as shown at the bottom of the page.

Some tedious manipulation reveals that the rank of $\mathbf{C}_{co}$ is only one, which means that the four curves are identical.

Thus, all of the four conics derived from the orthonormality of the rotation matrix are identical, and this means that the pose parameters cannot be uniquely determined by the rotation constraints. To extract the pose parameters from the reduced lidar essential matrix, we should utilize other independent constraints such as the plane orthogonality.

APPENDIX B

DERIVATION OF THE ROTATION PARAMETERS

A. Rotation Parameter r_{22} : From the orthonormality of $\mathbf{R}$,

$$\begin{aligned} r_{22} &= \pm \sqrt{1 - r_{12}^2 - r_{23}^2} \\ &= \pm \sqrt{1 - \lambda^2(v_1^2 + v_2^2)} = \pm \sqrt{1 - \lambda^2 k_0}. \end{aligned} \quad (81)$$

We can also obtain value of r_{22} using another equation $r_{21}^2 + r_{22}^2 + r_{23}^2 = 1$:

$$\begin{aligned} r_{22} &= \pm \sqrt{1 - r_{21}^2 - r_{23}^2} \\ &= \pm \sqrt{1 - \lambda^2(v_3^2 + v_4^2)} = \pm \sqrt{1 - \lambda^2 \hat{k}_0} \end{aligned} \quad (82)$$

where we define that $\hat{k}_0 \equiv v_3^2 + v_4^2$. The two constant k_0 and $\hat{k}_0$ should be identical, but actually not, because of the uncertainty in the linear estimation under some noise. We just use k_0 for simplicity in the whole of this paper.

$$\mathbf{C}_{co} = \begin{bmatrix} n_1^2 & 2n_1x_1 & v_3^2 + v_4^2 & x_1^2 - 1 \\ n_2^2 + n_3^2 & 2(n_2x_2 + n_3x_3) & v_1^2 & x_2^2 + x_3^2 - 1 \\ n_4^2 + n_5^2 & 2(n_4x_4 + n_5x_5) & v_2^2 & x_4^2 + x_5^2 - 1 \\ n_2n_4 + n_3n_5 & n_2x_4 + n_3x_5 + n_4x_2 + n_5x_3 & v_1v_2 & x_2x_4 + x_3x_5 \end{bmatrix}. \quad (80)$$

B. Rotation Parameters r_{11} , r_{33} , r_{31} , and r_{13} : From the orthogonality of $\mathbf{R}$, we obtain two equations

$$r_{11}^2 + r_{21}^2 + r_{31}^2 = 1 \quad (83)$$

$$r_{11}r_{12} + r_{21}r_{22} + r_{31}r_{32} = 0. \quad (84)$$

Multiplying both sides of (83) by r_{32}^2 ,

$$r_{11}^2 r_{32}^2 + r_{21}^2 r_{32}^2 + r_{31}^2 r_{32}^2 = r_{32}^2. \quad (85)$$

Third term of (85) is substituted by (84)

$$r_{11}^2 r_{32}^2 + r_{21}^2 r_{32}^2 + (r_{11}r_{12} + r_{21}r_{22})^2 = r_{32}^2. \quad (86)$$

Equation (86) can be regarded as a quadratic equation of r_{11} as

$$(r_{12}^2 + r_{32}^2)r_{11}^2 + (2r_{12}r_{21}r_{22})r_{11} + (r_{21}^2 r_{32}^2 + r_{21}^2 r_{22}^2 - r_{32}^2) = 0 \quad (87)$$

and by the quadratic formula

$$r_{11} = \frac{-r_{12}r_{21}r_{22}}{r_{12}^2 + r_{32}^2} \pm \frac{\sqrt{r_{12}^2 r_{21}^2 r_{22}^2 - (r_{12}^2 + r_{32}^2)(r_{21}^2 r_{32}^2 + r_{21}^2 r_{22}^2 - r_{32}^2)}}{r_{12}^2 + r_{32}^2}. \quad (88)$$

The discriminant of (88) can be simplified as

$$r_{12}^2 r_{21}^2 r_{22}^2 - (r_{12}^2 + r_{32}^2)(r_{21}^2 r_{32}^2 + r_{21}^2 r_{22}^2 - r_{32}^2) \quad (89)$$

$$= r_{32}^2(-r_{21}^2(r_{12}^2 + r_{22}^2 + r_{32}^2) + r_{12}^2 + r_{32}^2) \quad (90)$$

$$= r_{32}^2(-r_{21}^2 + r_{12}^2 + r_{32}^2) \quad (\because r_{12}^2 + r_{22}^2 + r_{32}^2 = 1) \quad (91)$$

$$= r_{23}^2 r_{32}^2. \quad (\because r_{12}^2 + r_{32}^2 = r_{21}^2 + r_{23}^2). \quad (92)$$

By putting (92) into (88), r_{11} becomes

$$\begin{aligned} r_{11} &= \frac{-r_{12}r_{21}r_{22} \pm r_{23}r_{32}}{r_{12}^2 + r_{32}^2} \\ &= \frac{-v_1 v_3 r_{22} \pm v_2 v_4}{v_1^2 + v_2^2} \\ &= -k_1 r_{22} \pm k_2. \end{aligned} \quad (93)$$

Similarly, other three elements are derived as

$$r_{33} = -k_2 r_{22} \pm k_1 \quad (94)$$

$$r_{31} = -k_3 r_{22} \mp k_4 \quad (95)$$

$$r_{13} = -k_4 r_{22} \mp k_3. \quad (96)$$

REFERENCES

- [1] D. F. Wolf and G. S. Sukhatme, "Mobile robot simultaneous localization and mapping in dynamic environments," *Auton. Robots*, vol. 19, no. 1, pp. 53–65, 2005.
- [2] P. Newman, D. Cole, and K. Ho, "Outdoor slam using visual appearance and laser ranging," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2006, pp. 1180–1187.
- [3] A. Nchter, K. Lingemann, J. Hertzberg, and H. Surmann, "6d slam with approximate data association," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2005, pp. 242–249.
- [4] D. M. Cole and P. M. Newman, "Using laser range data for 3d slam in outdoor environments," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2006, pp. 1556–1563.
- [5] S. Thrun, M. Montemerlo, H. Dahlkamp, D. Stavens, A. Aron, J. Diebel, P. Fong, J. Gale, M. Halpenny, G. Hoffmann, K. Lau, C. Oakley, M. Palatucci, V. Pratt, P. Stang, S. Strohband, C. Dupont, L.-E. Jendrosseck, C. Koelen, C. Markey, C. Rummel, J. van Niekerk, E. Jensen, P. Alessandrini, G. Bradski, B. Davies, S. Ettinger, A. Kaehler, A. Nefian, and P. Mahoney, "Stanley: The robot that won the DARPA grand challenge," *J. Field Robot.*, vol. 23, no. 9, pp. 661–692, 2006.
- [6] M. Montemerlo, J. Becker, S. Bhat, H. Dahlkamp, D. Dolgov, S. Ettinger, D. Haehnel, T. Hilden, G. Hoffmann, B. Huhnke, D. Johnston, S. Klumpp, D. Langer, A. Levandowski, J. Levinson, J. Marcil, D. Orenstein, J. Paefgen, I. Penny, A. Petrovskaya, M. Pfueger, G. Stanek, D. Stavens, A. Vogt, and S. Thrun, "Junior: The stanford entry in the urban challenge," *J. Field Robot.*, vol. 25, no. 9, pp. 569–597, 2008.
- [7] J. P. Fentanes, E. Zalama, and J. Gómez-García-Bermej, "Algorithm for efficient 3d reconstruction of outdoor environments using mobile robots," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2011, pp. 3275–3280.
- [8] Y. Bok, Y. Jeong, D.-G. Choi, and I. S. Kweon, "Capturing village-level heritages with a hand-held camera-laser fusion sensor," *Int. J. Comput. Vision*, vol. 94, no. 1, pp. 36–53, 2011.
- [9] Q. Zhang and R. Pless, "Extrinsic calibration of a camera and laser range finder (improves camera calibration)," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, vol. 3, 2004, pp. 2301–2306.
- [10] Y. Bok, D.-G. Choi, P. Vasseur, and I. S. Kweon, "Extrinsic calibration of non-overlapping camera-laser system using structured environment," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2014, pp. 436–443.
- [11] J.-E. Ha, "Extrinsic calibration of a camera and laser range finder using a new calibration structure of a plane with a triangular hole," *Int. J. Control, Autom. Syst.*, vol. 10, no. 6, pp. 1240–1244, 2012.
- [12] C. Gao and J. R. Spletzer, "On-line calibration of multiple lidars on a mobile vehicle platform," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2010, pp. 279–284.
- [13] J. Underwood, A. Hill, and S. Scheding, "Calibration of range sensor pose on mobile platforms," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2007, pp. 3866–3871.
- [14] M. Sheehan, A. Harrison, and P. Newman, "Automatic self-calibration of a full field-of-view 3d n-laser scanner," in *Experimental Robotics*. New York, NY, USA: Springer, 2014, pp. 165–178.
- [15] W. Maddern, A. Harrison, and P. Newman, "Lost in translation (and rotation): Rapid extrinsic calibration for 2d and 3d lidars," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2012, pp. 3096–3102.
- [16] J. Levinson and S. Thrun, "Unsupervised calibration for multi-beam lasers," in *Experimental Robotics*. New York, NY, USA: Springer, 2014, pp. 179–193.
- [17] M. He, H. Zhao, J. Cui, and H. Zha, "Calibration method for multiple 2d lidars system," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2014, pp. 3034–3041.
- [18] D.-G. Choi, Y. Bok, J.-S. Kim, and I. S. Kweon, "Extrinsic calibration of 2d laser sensors," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2014, pp. 3027–3033.
- [19] R. Molana and C. Geyer, "Motion estimation with essential and generalized essential matrices," in *Imaging Beyond the Pinhole Camera*. New York, NY, USA: Springer, 2006, pp. 107–123.
- [20] R. Pless, "Using many cameras as one," in *Proc. IEEE Conf. Comput. Vision Pattern Recog.*, vol. 2, 2003, pp. II–587.
- [21] D. W. Marquardt, "An algorithm for least-squares estimation of nonlinear parameters," *J. Soc. Ind. Appl. Math.*, vol. 11, no. 2, pp. 431–441, 1963.
- [22] P. J. Besl and N. D. McKay, "Method for registration of 3-d shapes," in *Robotics—DL Tentative*. Bellingham, WA, USA: SPIE, 1992, pp. 586–606.



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